MSC CYBERSECURITY & DATA SCIENCE · ESAIP ANGERS · GROUP 2

# TrustNav

## Can a fleet believe its satellite position?

A trust score from 0 to 100 for every vehicle, every second. Explained in plain words. Acted on before a lie reaches dispatch.




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Tutor: Sofiane Hamrioui · Oral examination, 13 November 2026

THE PROBLEM

# Everything that moves trusts GNSS blindly

**1 500**

flights spoofed per day around the Baltic and conflict zones, summer 2024. A daily event for aviation since 2022.



**Nantes airport, 2017**
A jammer left running in a parked car delayed several flights. Fine: €2 000.



**Lyon airport, 2023**
A truck driver's jammer, used to escape fleet tracking, disrupted signals until the ANFR traced it.

WHAT EXISTS TODAY

**Fleet telematics**
shows whatever position it is given

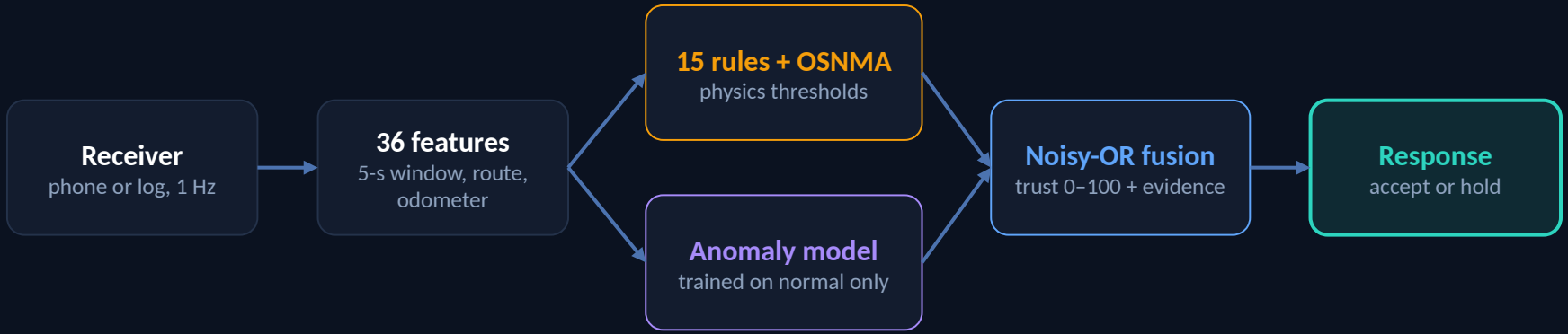
**Receiver flags**
a state code, no explanation

**Fixed probes, anti-jam antennas**
protect a site, not a moving van

**The gap** Nobody tells the operator, per vehicle and per second, whether the position is true, or acts on it.

THE IDEA

# A software layer beside the receiver




## Scores

0 to 100, per vehicle and drone, every second

## Explains

in plain words, with values: “Fix moved 7 998 m in 1 s (limit 250 m)”

## Responds

refuses the fix and tracks the vehicle by its odometer along its route

DETECTION

# Three witnesses, one verdict

## Rules

15 physical checks with named thresholds

**CATCHES** a van cannot jump 250 m in a second

**MISSES** slow attacks never cross a threshold

*60 m/s · 8 m/s² · 6 dB-Hz C/N0 shift · PDOP > 5*

## Anomaly model

Isolation Forest + tail detector, trained on normal driving only

**CATCHES** a fix 90 m off the road with clean signals

**MISSES** noisier; must be calibrated on the receiver

*risk = 25 · (−log10 p): 50 is a 1-in-100 event*

## Galileo OSNMA

satellites sign their navigation data (operational since July 2025)

**CATCHES** forged data: the signature fails

**MISSES** relay attacks keep it valid; GPS not covered

*one witness among three, not a defence*

## Fused by noisy-OR, not averaged

risk = 100 · [1 − (1 − 0.6 · p rules)(1 − 0.8 · p model)]  trust = 100 − risk

MODEL SAYS 90
RULES SAY 20

55 average dilutes it

75 noisy-OR keeps it

# The response layer: the lie stops here

1

## ACCEPT

trust at least 50 and no critical rule: the fix passes through unchanged.

2

## HOLD

the fix is refused. The position handed on is dead-reckoned from the last trusted fix with the odometer, along the planned route. A spoofer can touch neither.

3

## RELEASE

after 5 clean seconds and a fix within 60 m + 3 % of the distance travelled. After 60 s the bound is ten times looser; a fix still kilometres away stays refused.

CONTINUE VERIFY ISOLATE




*The console during the teleport: the van's real path (left), the reported position 8 km away (right), and the dashed lie between them.*




### What the operator reads

Score 15, Untrusted. Rules 45, model 100, combined risk 85.

*"The receiver places the van 8.0 km from the road it is actually on… TrustNav has stopped passing this position on."*

EVALUATION

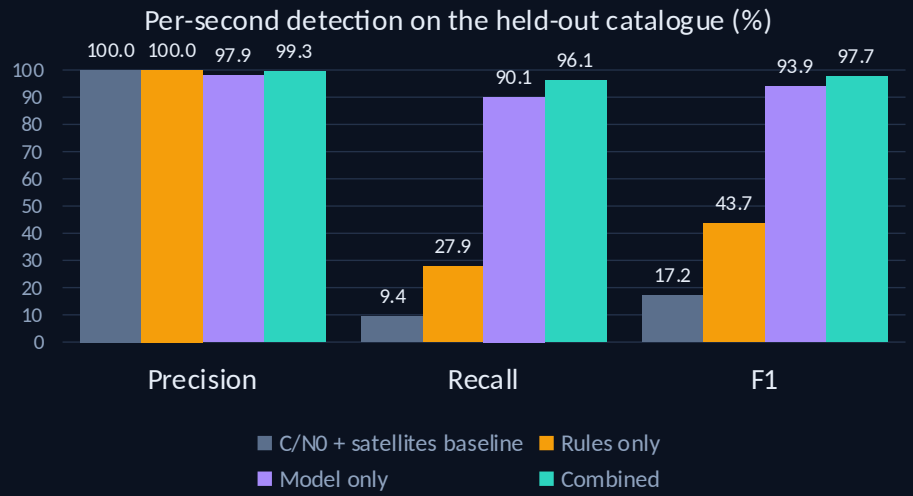
# Tested on routes it has never seen

**99.3 %**
PRECISION

**96.1 %**
RECALL

**0.44 %**
FALSE ALARMS

**1 s**
MEDIAN DELAY



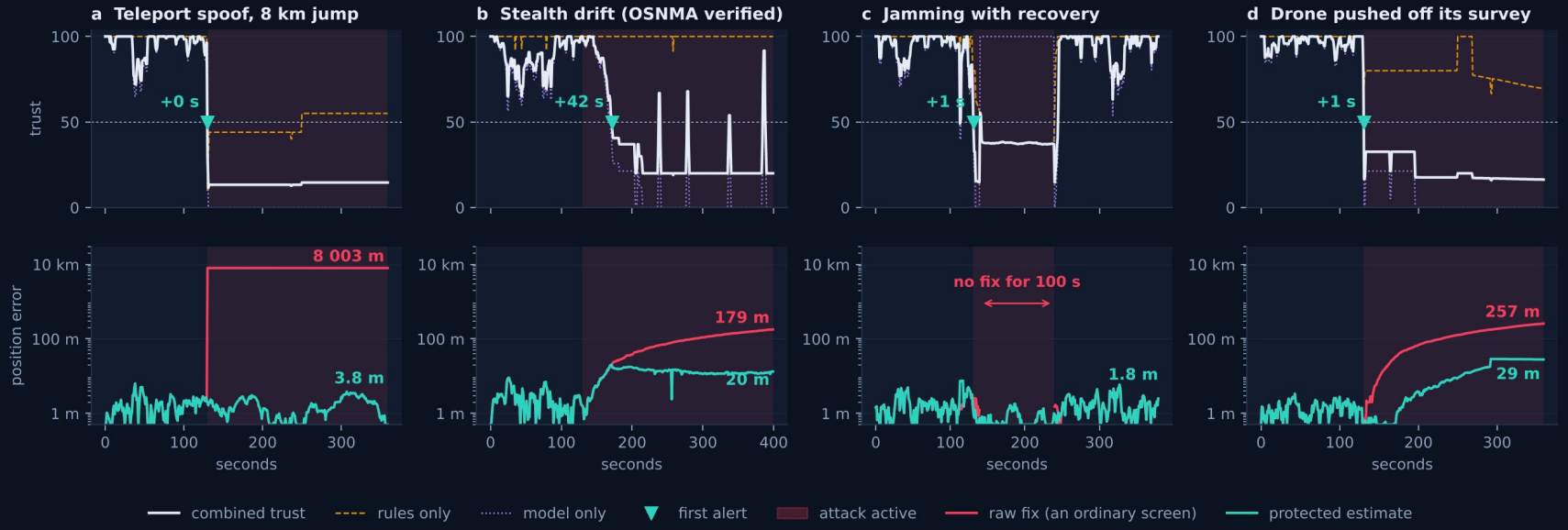

**10 scenarios · 3 800 seconds**
1 550 under attack, 7 attack types, real Angers streets

**One role per route**
train, validate or test: never shared

**95 % intervals over scenarios**
precision 97.9–100 · recall 89.4–99.6

FOUR HELD-OUT ATTACKS

# What happens, second by second



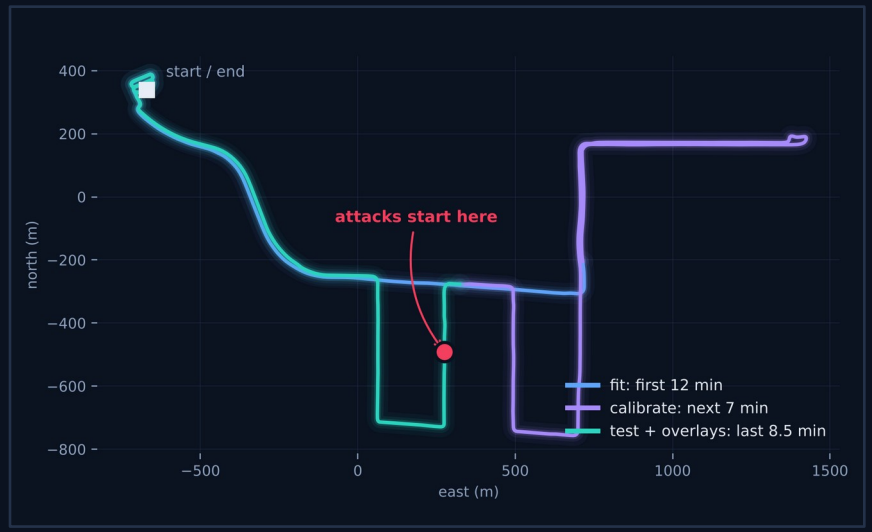

**7 / 7**
attack episodes detected

**8 km → 29 m**
worst position handed on

**100 s → 0 s**
blank screen under jamming

**42 s**
stealth drift caught, 12 m off the road

# Then we tried a real receiver




*Public recording: u-blox ZED-F9K with a NovAtel SPAN reference, Mountain View, 27.6 min (Google gps-measurement-tools, Apache-2.0). The reference track serves as route and odometer, so protected errors are idealised.*

**0** — **false alarms from the rules**
in the whole drive (minimum trust 90). The classic signal baseline raised 86 s.

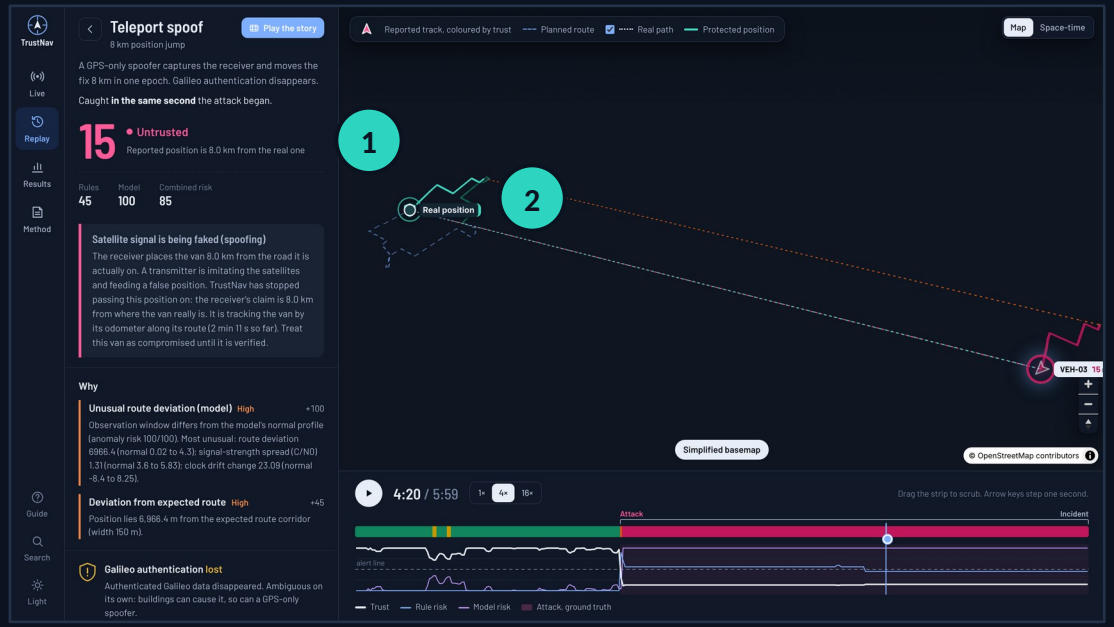
**89 %** — **of seconds alerted by the simulator-trained model**
so it must be re-calibrated on the receiver in use. Re-calibrated: no alert in the last 8.5 min.

**0 · 35 · 1 s** — **to catch overlaid teleport, drift, jamming**
protected estimate within 0.8, 24 and 0.8 m of the reference.

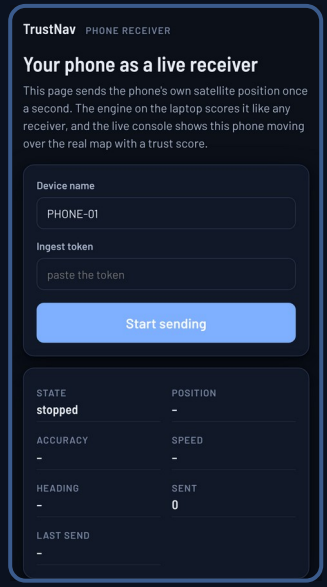
**Real data found two weaknesses. Both fixed.**
Tail scores now in log space · the 60-s release is bounded

LIVE DEMO

# The console, and a phone as a receiver




3




**1** score, verdict, evidence **2** real path vs the reported lie **3** your phone, scored live, with an attack switch

trustnav.vercel.app · open source · Python engine, 32 tests · Supabase, row-level security · Next.js console, 109 end-to-end checks

HONEST CLAIMS

# What we claim, what we don't, what's next

## Limits

- one simulator: the numbers are an upper bound
- route checks need an assigned route
- the response is as good as its odometer
- OSNMA simulated; no multi-antenna evidence

## Safety and ethics

- nothing is transmitted: attacks are software
- future hardware is receive-only
- GDPR: purpose, 3-day retention, transparency
- a research indicator, not certified integrity

## Next

- hours of real drives in Angers
- replay Jammertest 2023 and TEXBAT captures
- a second receiver against synchronous spoofing
- real OSNMA from raw Galileo pages

*A per-second, explained trust score, and a response that keeps a lie from the systems that act on it.*

# Questions?

trustnav.vercel.app · github.com/domail2karandeshmukh-cyber/trustnav